

**DUAL STEGANOGRAPHY FOR HIDING TEXT
IN VIDEO BY LINKED LIST METHOD**

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Theoretical

The present improvement in computational power and advancement has prompted the necessity for particularly secured data correspondence. Exceptional contrasted with different frameworks for secure correspondence is Steganography-a covert composed work. It is a specialty of disguising the very nearness of passed on message itself..

The fact of the matter is to layout a steganography estimation which cover the message behind the photo and in addition give more security than others .another steganography system for embeddings both substance or picture in cover pictures by using LSB and Link List method is realized. This steganography method is absolutely a period space (pixel based) and secret messages are embedded clearly into 24-bit shading picture.

Two ways are suited embeddings the riddle data inside cover picture, for instance, progressive encoding and self-assertive encoding for both substance and picture .For the purpose behind security, encryption technique is used with a userdefined key. RGB picture sort out is used to upgrade the idea of the stego picture. At last that RGB picture will saved as BMP picture record so no lossy weight can happen and the principal message don't pulverize and can be separate as it is by all accounts.

Perspective extent for both substance and picture ensuing to stowing without end in cover picture keeps up definitely same. Execution of proposed steganography technique is evaluated by finding out estimations of MSE(Mean square screw up), PSNR(Peak banner to upheaval extent), ET(Elapse time). Twofold Steganography is the route toward using Steganography joined with Cryptography.

Steganography is the path toward stowing without end mystery data's in the media reports, for instance, sound, pictures, chronicles et cetera. Cryptography is a branch of number-crunching stressed over the examination of concealing and revealing information and besides to demonstrate production of messages. In this paper, Dual Steganography thought has been associated with secure the principal chronicles from unapproved person.

The methodology has been done by embeddings the principal video inside another video. Both the chronicles are changed over into diagrams first. By then the individual edges of one of a kind video are tried with the edges of another video. Resulting to completing the testing strategy, the yield traces are solidified to get the mixed video.

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CHAPTER-1

INTRODUCTION

Steganography is the strength of elusive correspondence which covers any private information inside a honest looking spread test. The word Steganography is gotten from the Greek words "stegos" signifying "cover" and "grafia" hinting "communicating" depicting it as "secured formed work". Steganography is a data camouflaging procedure made beginning late. It is a strategy that grips visual or aural emphasis to front line sight and sound, and that presents the enigma data when all is said in done society media to exchange modernized media passing on private data to complete secret correspondences. Steganography is remarkable in association with cryptography. The objective of cryptography is to give secure trades by changing the information into an edge that can't be gotten on. Steganography strategies, then again, cover the closeness of the message itself, which makes it cumbersome for a third individual to discover the message. Not at all like steganography, sending blended data may draw thought. As necessities be, cryptography is not the enormous reaction for secure correspondence yet rather just piece of the game-plan. The two structures can be utilized together to better ensure data. Essential steganography demonstrate is appeared in

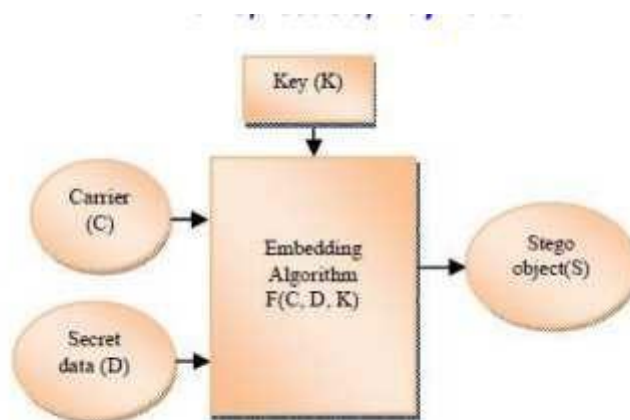


Fig.1.1

Fig 1.1: Basic steganography model

The major model involves Carrier(C), Secret Data(D), Stego Key(K).

1. Transporter is the cover challenge in which the message is embedded.
2. Puzzle data can be any sort of mystery data that can be plain substance, figure content or other picture.

3. Key generally used to ensure that elite recipient having the unwinding key will have the ability to isolate the message from a cover-dissent.

4. By using the embedding estimation, the riddle data is embedded into the cover question in a way that does not change the primary picture in a human perceptible manner.

5. At long last, the stego protest which is the yield of the procedure is the cover-question with the furtively inserted data.

So far Cryptography is utilized as a bit of many structures yet utilizing it with Audio records is another Stronger Techniques. The technique of Cryptography occurs with Audio File for exchanging more secure delicate information. The Sensitive Data is Encoded with a Video File and Passed over Insecure Channels to flip side of Systems. Here we are utilizing .wav report Format for Encryption and Decryption of Message.

The given message will be encoded with a given video record utilizing a conundrum key. The System will then install the enigma message into the video file. The outcome will be another video record, which has the conundrum message in it. While unraveling a similar key ought to be given for blended video answer to get the astound message from it.

Main Objective:

Stenography with Video is a desktop application. The purpose behind this application is to give the security to the mystery information. This application doesn't empower the software engineers to see the data, can see just video record when it is being disregarded the web. By then at the recipient side the primary information i.e., plain substance will be expelled from the sound by performing translating operations.

1.1.PROJECT ARCHITECTURE

The general architecture consists of two phases:

1. Hiding data in Video (Encryption)
2. Retrieval of original information (Decryption)

A. Overall Design For Encryption

In Encryption designing, first the cover video is changed over into a progression of housings by removing them. Each evacuated packaging addresses a photo. By then the transmit information which must be embedded inside the video report is first mixed using Feistel arrange with different keys ($K_0, K_1, K_2 \dots K_n$). The mixed information is then confined into bytes of data. By then, every byte of data is embedded

into each video diagram in a gathering using Linked List Structure Message Embedding Technique. In the wake of introducing the information into diagrams, a gathering of Stego Frames will be gotten. The embedded edges are called as Stego Frame. Later the Stego Housings are solidified to get the Stego Video containing the hid message inside.

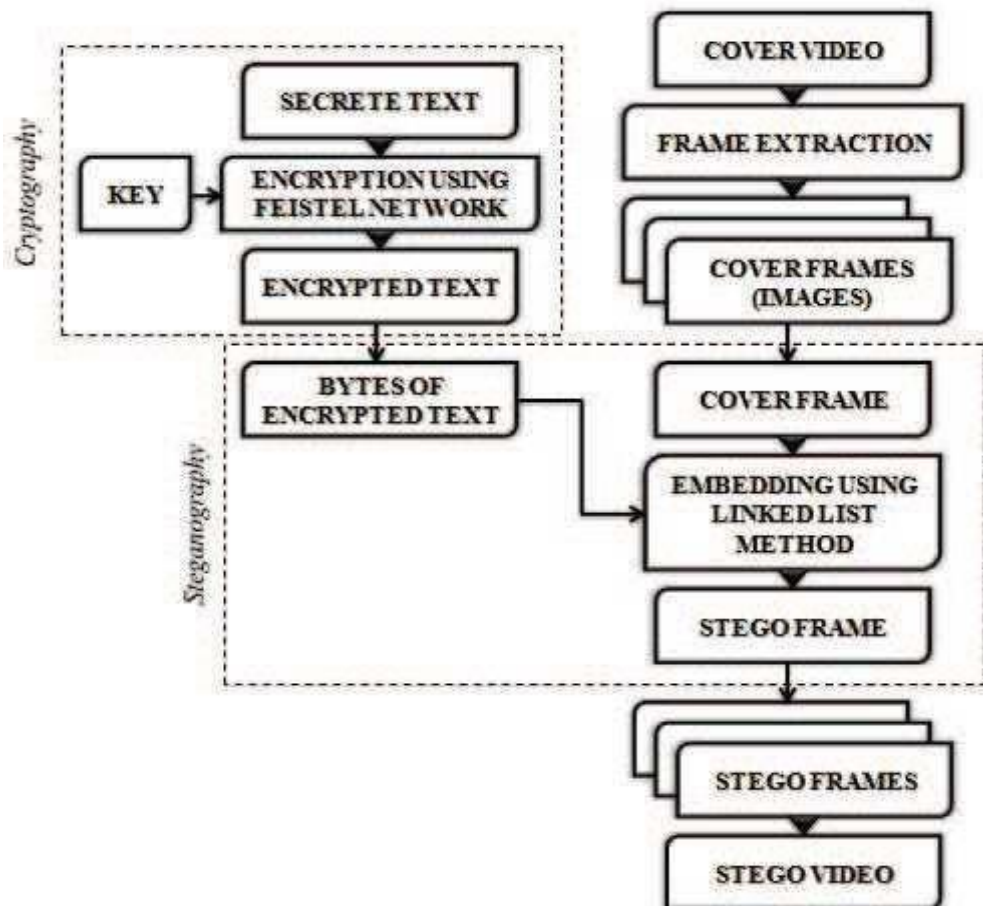


Fig: Encryption Architecture

B. Overall Design For Decryption:

In Decryption Architecture Fig 2, first the Stego Video containing disguised message is changed over into a progression of Stego plots by isolating them. Each removed diagram addresses a Stego picture. By then the release information is removed from Stego diagrams using Linked List Structure system. The evacuated substance will be as encoded message. The message is then decoded using Feistel Network with various keys

(K0, K1, K2 ... Kn) and the main message is obtained.

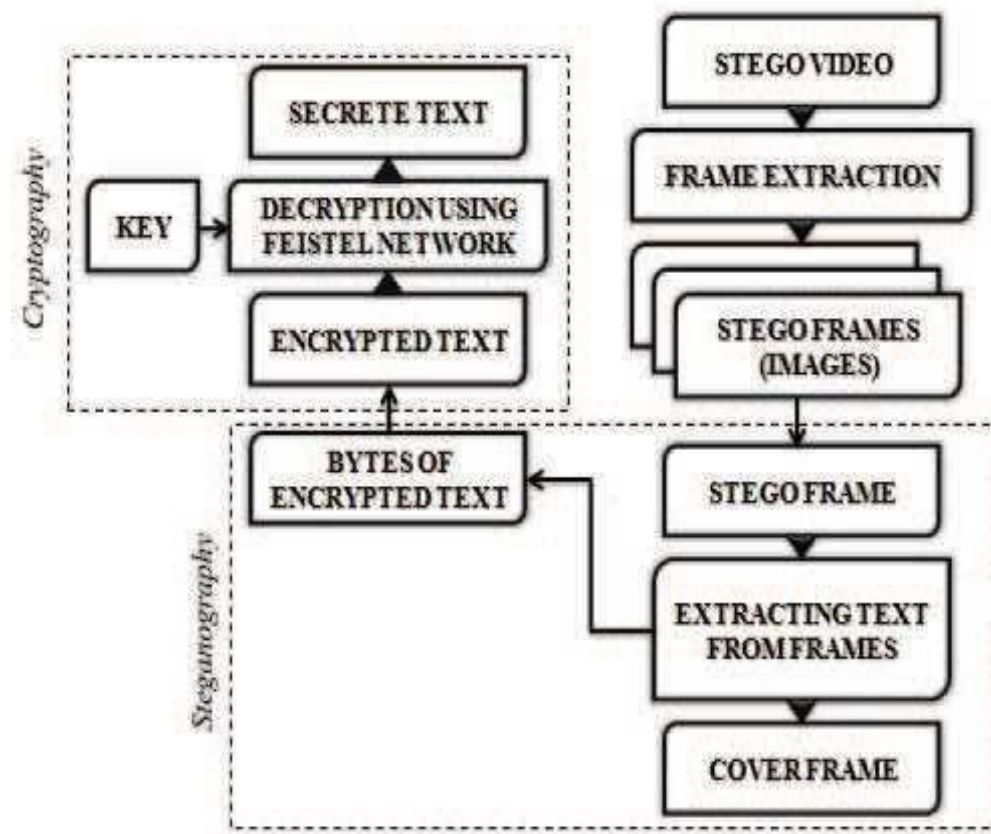


Fig: Decryption Architecture

1.2 MODULE DESCRIPTION

The modules are: Encryption process Decryption process

A. Module 1: Encryption Process

The methods related with encryption process are:

Isolating layouts from video

Encoding data using Feistel mastermind figuring

Embeddings content inside picture plots

Getting Stego video

1) Extracting diagrams from video

The original video (cover video) is changed over into a gathering of housings. Each packaging addresses a photo.

2) Encrypting data using Feistel sort out count

The transmit information is encoded using Feistel sort out count. In cryptography, a Feistel figure is a symmetric structure used as a piece of the improvement of square figures, named after the German IBM cryptographer Horst

Feistel; it is in like manner ordinarily known as a Feistel orchestrate. A significant course of action of square figures use the arrangement, including the

Data Encryption Standard (DES).

The Feistel structure has the favored point of view that encryption and unscrambling operations are generally the same as, even undefined now and again, requiring only a reversal of the key timetable. In this manner the measure of the code or equipment required to realize such a figure is practically part. Feistel advancement is iterative in nature which makes executing the cryptosystem in hardware less requesting

CHAPTER-3

STUDY OF THE SYSTEM

3.1FEASIBILITY STUDY

The practicality of the attempt is broke down in this stage and business recommendation is Put forward with a to an awesome degree wide course of action for the task and some cost gages. Amidst framework examination the probability examination of the proposed structure is to be done. This is to guarantee that the proposed structure is not a weight to the affiliation. For likelihood examination, some comprehension of the enormous necessities for the framework is basic.

Three key considerations related with the reasonableness examination are

- “ ECONOMICAL FEASIBILITY
- “ TECHNICAL FEASIBILITY
- “ SOCIAL FEASIBILITY

ECONOMICAL FEASIBILITY

This examination is done to check the money related effect that the structure will have on the connection. The measure of hold that the affiliation can fill the imaginative work of the framework is restricted. The uses must be upheld. Along these lines the made framework as Ill inside the money related game plan and this was refined in light of the way that a huge piece of the advancements utilized are uninhibitedly open. Just the patch up things must be obtained.

TECHNICAL FEASIBILITY

This examination is done to check the specific good judgment, that is, the particular necessities of the framework. Any framework influenced must to not have an omnipresence on the accessible specific assets. This will prompt levels of commonness on the open particular assets. This will impel levels of predominance being resolved to the customer.

The made framework must have an unassuming fundamental, as just superfluous or invalid changes are required for executing this structure.

SOCIAL FEASIBILITY

The bit of study is to check the level of assertion of the framework by the client. This joins the way toward setting up the client to utilize the structure effectively. The client must not feel undermined by the structure, rather should remember it as a need. The level of confirmation by the clients just relies on the techniques that are utilized to instruct the client about the structure and to make him comfortable with it. His level of confirmation must be raised with the target that he is also arranged to make some supportive info, which is Income, as he is the last client of the structure.

CHAPTER-4

SYSTEM SPECIFICATIONS

3.1. HARDWARE SPECIFICATIONS

Processor: Pentium-III (or) Higher

Ram: 64MB (or) Higher

Cache: 512MB

Hard disk: 10GB

3.2. SOFTWARE SPECIFICATIONS

Tools: Micro Soft Front

Operating System: Windows NT/2000

Languages: Java 2 Standard Edition

CHAPTER-4

SOFTWARE ENVIRONMENT

4.1 JAVA BASICS

Java technology is both a programming language and a platform.

4.1.1 The Java Programming Language:

The Java programming dialect is an abnormal state dialect that can be described by the greater part of the accompanying popular expressions:

Basic

Engineering impartial Object situated Portable

Dispersed

Superior Interpreted Multithreaded Robust

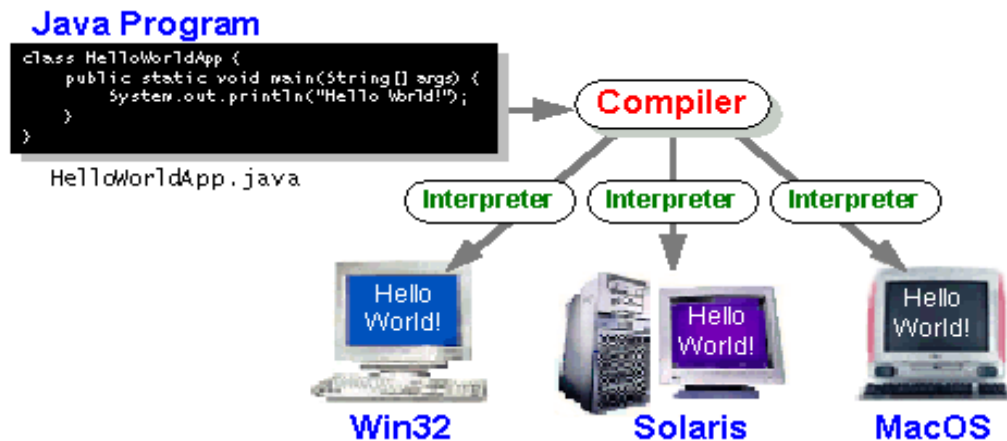
Dynamic

Secure

With most programming lingos, you either join or translate a program so you can run it on your PC. The Java programming vernacular is astonishing in that a program is both assembled and deciphered. With the compiler, first you make an understanding of a program into a widely appealing vernacular called Java byte codes — the stage self-governing codes deciphered by the go between on the Java arrange. The go between parses and runs each Java byte code heading on the PC. Gathering happens just once; interpretation happens each time the program is executed. The going with figure speaks to how this capacities.

You can consider Java byte codes as the machine code rules for the Java Virtual Machine (Java VM). Every Java arbiter, paying little mind to whether it's a change instrument or a Web program that can run applets, is an utilization of the Java VM. Java byte codes empower make "to create once, run wherever" possible. You can arrange your program into byte codes on any phase that has a Java compiler. The byte codes would then have the capacity to be continue running on any execution of the Java VM. That infers that as long as a PC has a Java VM, a comparable program written in the Java programming vernacular can continue running on

Windows 2000, a Solaris workstation, or on an iMac.



4.1.2 The Java Platform:

A phase is the hardware or programming condition in which a program runs. I've quite recently said presumably the most understood stages like Windows 2000, Linux, Solaris, and MacOS. Most stages can be depicted as a mix of the working system and hardware. The Java arrange contrasts from most extraordinary stages in that it's an item simply organize that continues running over other gear based stages.

The Java organize has two sections:

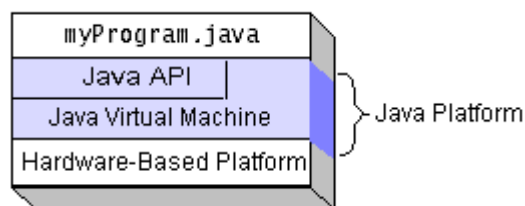
The Java Virtual Machine (Java VM)

The Java Application Programming Interface (Java API)

You've quite recently been familiar with the Java VM. It's the base for the Java arrange and is ported onto diverse gear based stages.

The Java API is a broad social occasion of moment programming sections that give various supportive limits, for instance, graphical UI (GUI) contraptions. The Java API is amassed into libraries of related classes and interfaces; these libraries are known as groups. The accompanying fragment, What Can Java Technology Do? Highlights what value a part of the groups in the Java API give.

The going with figure depicts a program that is running on the Java organize. As the figure appears, the Java API and the virtual machine protect the program from the equipment .



Adjacent code can't abstain from being code that after you mastermind it, the amassed code keeps running on a particular apparatus compose. As a stage self-decision condition, the Java stage can be a bit slower than neighborhood code. In any case, sharp compilers, all around tuned mediators, and finally byte code compilers can pass on execution near that of close-by code without debilitating littleness.

What Can Java Technology Do?

The most extensively saw sorts of endeavors written in the Java programming tongue are applets and applications. In the event that you've surfed the Web, you're without a doubt enough okay with applets. An applet is a program that holds fast to specific traditions that enable it to keep running inside a Java-connected with program.

In any case, the Java programming dialect is not only to create bewildering, associating with applets for the Web. The exhaustively profitable, odd state Java programming vernacular is besides an effective programming stage. Utilizing the liberal API, you can make many manages of undertakings.

An application is a free program that runs coordinate on the Java orchestrate. A unique of utilization known as a server serves and bolster customers on a

organize. Occasions of servers are Web servers, center individual servers, mail servers, and print servers. Another particular program is a servlet. A servlet can in every practical sense be thought of as an applet that keeps running on the server side. Java Servlets are a perceptible decision for building brilliant web applications, supplanting the utilization of CGI substance. Servlets take after applets in that they are runtime advancements of employments. Rather than working in programs, in any case, servlets keep running inside Java Web servers, sorting out or fitting the server. How does the API bolster every last one of these sorts of endeavors? It does everything considered with gatherings of programming

segments that gives an expansive collection of support. Each full execution of the Java compose gives you the running with highlights:

The essentials: Objects, strings, strings, numbers, data and yield, data structures, system properties, date and time, and so forth.

Applets: The game plan of customs used by applets.

Systems association: URLs, TCP (Transmission Control Protocol), UDP (User Data gram Protocol) associations, and IP (Internet Protocol) addresses.

Internationalization: Help for making programs that can be headed for clients around the world. Endeavors can along these lines change as per particular territories and be showed up in the most ideal vernacular.

Security: Both low level and peculiar state, including electronic engravings, open and private key association, get the chance to control, and certifications.

Programming areas: Known as Java BeansTM, can connect with existing part designs.

Question serialization: Allows lightweight confirmation and correspondence by strategies for

Remote Method Invocation (RMI).

Java Database Connectivity (JDBCTM): Provides uniform access to a wide arrangement of social databases.

The Java orchestrate additionally has APIs for 2D and 3D diagrams, responsiveness, servers, joint effort, correspondence, talk, improvement, and that is only a look at a bigger issue. The running with figure delineates what is combined into the Java 2 SDK.

By what means Will Java Technology Change My Life?

I can't guarantee you pervasiveness, fortune, or even an employment on the off chance that you take in the Java programming vernacular. In any case, it is apparently going to improve your endeavors and requires less exertion than different vernaculars. I accept that Java advancement will engage you to do the running with:

Begin rapidly: Although the Java programming vernacular is a remarkable test masterminded tongue, it's certainly not hard to learn, particularly to programme builds formally OK with C or C++.

Influence less to code: Comparisons of program estimations (class checks, system tallies, and so forth) endorse that a program written in the Java programming dialect can be four times more minor than a similar program in C++.

Shape better code: The Java programming vernacular empowers extraordinary coding practices, and its trash gathering engages you keep up a key division from memory spills. Its dispute introduction, its JavaBeans piece building, and its far reaching, effectively extendible API let you reuse other individuals' endeavored code and present less bugs.

Make programs all the more rapidly: Your change time might be as much as twice as smart instead of shaping a relative program in C++. Why? You influence less lines to out of code and it is a less requesting programming vernacular than C++.

Avoid organize conditions with 100% Pure Java: You can keep your program adaptable by keeping up an indispensable partition from the use of libraries written in different languages. The 100% Pure Java™ Product Certification Program has a store of evident process manuals, white papers, presents, and essentially indistinguishable materials on the web.

Frame once, run wherever: Because 100% Pure Java programs are gathered into machine-free byte codes, they run reliably on any Java mastermind.

Dissipate programming all the more effectively: You can refresh applets effectively from a focal server. Applets manhandle the segment of enabling new classes to be stacked "on the fly,"

ODBC CONNECTIVITY

Microsoft Open Database Connectivity (ODBC) is a standard programming interface for application organizers and database structures suppliers. Before ODBC changed into a recognized standard for Windows undertakings to interface with database structures, engineers anticipated that would utilize specific dialects for every database they anticipated that would associate with. Before long, ODBC has settled on the decision of the database framework basically unessential from a coding viewpoint, which is as it ought to be. Application engineers have liberally more fundamental things to stress over than the sentence structure that is required to port their program starting with one database then onto the accompanying when business needs out of the blue change.

Through the ODBC Administrator in Control Panel, you can demonstrate the particular database that is connected with a data source that an ODBC application program is formed to use. Consider an ODBC data source as a portal with a name on it. Each portal will lead you to a particular database. For example, the data source named Sales Figures might be a SQL Server database, however the Accounts Payable data source could imply an Access database. The physical database insinuated by a data source can harp wherever on the LAN.

The ODBC system archives are not presented on your structure by Windows 95. Or, on the other hand perhaps, they are presented when you setup an alternate database application, for instance, SQL Server Client or Visual Basic 4.0. Exactly when the ODBC image is presented in Control Panel, it uses a record called ODBCINST.DLL. It is in like manner possible to coordinate your ODBC data sources through a stay lone program called ODBCADM.EXE. There is a 16-bit and a 32-bit version of this program and every keep up an alternate summary of ODBC data sources.

From a programming point of view, the flawlessness of ODBC is that the application can be framed to utilize a near strategy of restrict calls to interface with any information source, paying little identity to the database merchant. The source code of the application doesn't change whether it talks with Oracle or SQL Server. I just demonstrate these two for instance. There are ODBC drivers open for two or three dozen praised database frameworks. Without a doubt, even Excel spreadsheets and plain substance records can be changed into information sources. The working structure utilizes the Registry data shaped

by ODBC Administrator to comprehend which low-level ODBC drivers are depended upon to talk with the information source, (for example, the interface to Oracle or SQL Server). The stacking of the ODBC drivers is clear to the ODBC application program. In a customer/server condition, the ODBC API even handles an imperative number of the system issues for the application programming engineer.

The upsides of this game plan are different to the point that you are no uncertainty sense there must be some catch. The essential downside of ODBC is that it isn't as competent as talking direct to the area database interface. ODBC has had different depreciators make the charge that it is too much immediate. Microsoft has always guaranteed that the fundamental factor in execution is the possibility of the driver programming that is utilized. As we might need to think, this is generous. The accessibility of good ODBC drivers has updated a surprising blueprint beginning late. Furthermore, at any rate, the feedback about execution is to some degree essentially corresponding to the general population who said that compilers could never organize the speed of unadulterated low level enrolling fabricate. Possibly not, yet rather the compiler (or ODBC) enables you to make cleaner programs, which construes you complete sooner. In the mean time, PCs get quicker dependably.

JDBC Connectivity

With an extreme target to set a free database standard API for Java; Sun Microsystems made Java Database Connectivity, or JDBC. JDBC offers a non specific SQL database get to system that gives a strong interface to a blend of RDBMSs. This strong interface is master using "module" database availability modules, or drivers. In the event that a database merchant wishes to have JDBC support, he or she should give the driver to each stage that the database and Java keep running on.

To get a more wide assertion of JDBC, Sun build up JDBC's structure concerning ODBC. As you found before in this portion, ODBC has wide help on a gathering of stages. Creating JDBC in light of ODBC will enable dealers to set up JDBC drivers accessible to be acquired to general society basically speedier than working up a completely new framework strategy.

JDBC was represented in March of 1996. It was discharged for a 90 day open examination that finished June 8, 1996. In light of client input, the last JDBC v1.0 particular was discharged a little while later.

The straggling leftovers of this area will cover enough data about JDBC for you to recognize what genuinely matters to it and how to utilize it adequately. This is in no way, shape or form, an entire review of JDBC. That would fill a whole book.

JDBC Aims:

Hardly any thing bundles are orchestrated without focuses at the most elevated purpose of the need list. JDBC is one that, due to its different objectives, drove the difference in the API. These goals, in conjunction with early correspondent input, have finished the JDBC class library into a strong structure for building database applications in Java.

The goals that were set for JDBC are critical. They will give you some data concerning why certain classes and functionalities bear in travel they do. The eight arrangement goals for JDBC are according to the accompanying:

1. SQL Level API

The makers felt that their essential target was to portray a SQL interface for Java. Regardless of the way that not the most insignificant database interface level possible, it is at an adequately low level for more lifted sum instruments and APIs to be made. Then again, it is at an adequately abnormal state for application programming specialists to use it undeniably. Fulfilling this target considers future gadget venders to "make" JDBC code and to hide a significant part of JDBC's complexities from the end customer.

2. SQL Conformance

SQL language structure changes as you move from database seller to database merchant. With an end goal to help a wide assortment of merchants, JDBC will enable any question proclamation to be gone through it to the fundamental database driver. This enables the availability module to deal with non-standard usefulness in a way that is reasonable for its clients.

3. JDBC must be implemental over basic database interfaces

The JDBC SQL API must "sit" over other normal SQL level APIs. This objective enables JDBC to utilize existing ODBC level drivers by the utilization of a product interface. This interface would make an interpretation of JDBC calls to ODBC and the other way around.

4. Give a Java interface that is solid with whatever is left of the Java structure

Because of Java's affirmation in the customer assemble until now, the makers feel that they should not stray from the present diagram of the middle Java structure.

5. Keep it clear

This target apparently appears in all item design target postings. JDBC is no exclusion. Sun felt that the arrangement of JDBC should be incredibly fundamental, considering only a solitary procedure for completing an errand for every part. Allowing duplicate convenience just serves to overwhelm the customers of the API.

6. Use strong, static written work wherever possible

Strong written work thinks about more botch checking to be done at collect time; also, less error appear at runtime.

7. Keep the normal cases direct

Since when in doubt, the average SQL calls used by the designer are direct SELECT's, INSERT's, DELETE's and UPDATE's, these inquiries should be anything but difficult to perform with JDBC. In any case, more personality boggling SQL declarations should moreover be possible.

Finally, I continued the use using Java Networking. Likewise, for logically reviving the store table i go for MS Access database.

Java has two things:

1. Programming Language.

2. Stage.

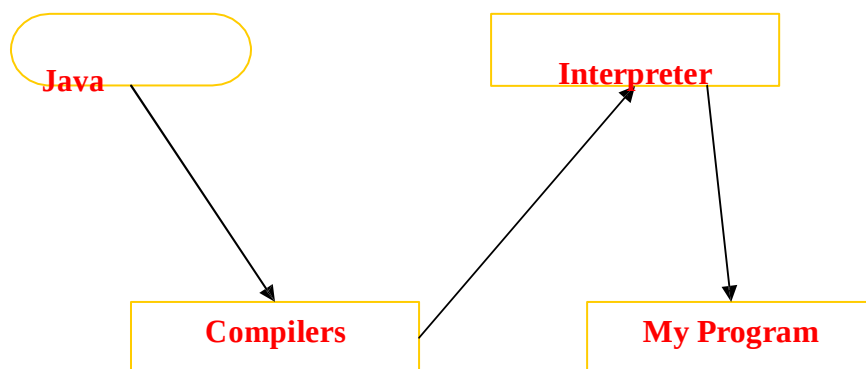
Java is an anomalous state programming lingo that is most of the going with: Simple Architecture-fair-minded Object-Oriented Portable

Coursed High Performance Interpreted

Multi hung Robust

Dynamic Secure

Java is furthermore astounding in that each Java program is both organized and interpreted. With a request you disentangle a Java program i



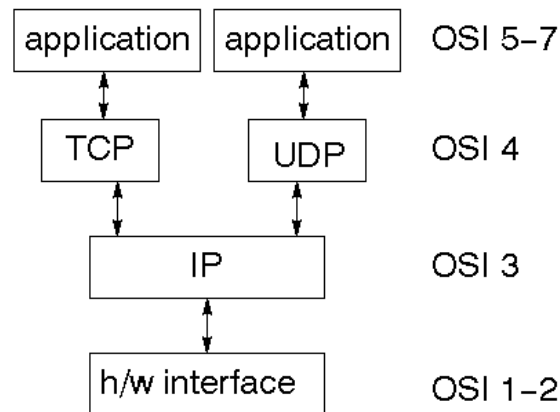
You can consider Java byte codes as the machine code rules for the Java Virtual Machine (Java VM). Every Java interpreter, paying little respect to whether it's a Java progression gadget or a Web program that can run Java applets, is an utilization of the Java VM. The Java VM can in like manner be completed in hardware.

Java byte codes empower make "to form once, run wherever" possible. You can organize your Java program into byte codes on my phase that has a Java compiler. The byte codes would then have the capacity to be run any use of the Java VM. For example, a comparative Java program can run Windows NT, Solaris, and Macintosh.

Networking:

TCP/IP Stack

The TCP/IP stack is shorter than OSI one:



TCP is a connection-oriented protocol; UDP (User Datagram Protocol) is a connectionless protocol.

IP datagram's

The IP layer gives a connectionless and inconsistent transport system. It considers each datagram self-governingly of the others. Any connection between datagram must be given by the higher layers. The IP layer supplies a checksum that fuses its own specific header. The header fuses the source and objective locations. The IP layer handles coordinating through an Internet. It is similarly responsible for isolating tremendous datagram into tinier ones for transmission and reassembling them at the other

end

UDP

UDP is also connectionless and conflicting. What it adds to IP is a checksum

for the substance of the datagram and port numbers. These are used to give a client/server illustrate - see later.

TCP

TCP supplies strategy for thinking to give a solid alliance engineered convention above IP. It

gives a virtual circuit that two methods can use to present.

Web addresses

Recollecting the genuine goal to utilize an association, you should be able to discover it. The Internet utilizes an address plot for machines with the target that they can be found. The address is a 32 bit number which gives the IP address. This encodes a system ID and all the all the additionally tending to. The system ID falls into different classes as appeared by the navigate of the structure address.

Structure address

Class A utilizes 8 bits for the structure address with 24 bits left finished for other tending to. Class B utilizes 16 bit orchestrate tending to. Class C utilizes 24 bit mastermind tending to and class D utilizes each of the 32.

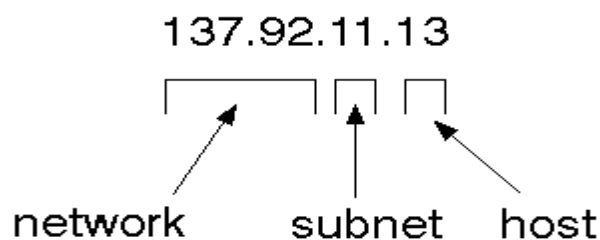
Subnet address

Inside, the UNIX coordinate is separated into sub systems. Building 11 is beginning at now on one sub system and uses 10-bit tending to, permitting 1024 phenomenal hosts.

Host address

8 bits are at long last utilized for have addresses inside our subnet. This places a point of confinement of 256 machines that can be on the subnet.

Total address



The 32 bit address is usually written as 4 digits separated by dots.

Port addresses

An organization exists on a host, and is perceived by its port. This is a 16 bit number. To establish a connection on a server, you send it to the port for that organization of the host that it is running on. This is not range straightforwardness! Beyond any doubt of these ports are "remarkable".

Connections

A connection is a data structure kept up by the system to manage orchestrate affiliations. A connection is made using the call connection. It reestablishes an entire number that looks like a record descriptor. Frankly, under Windows, this handle can be used with Read File and Write File limits.

```
#include <sys/types.h>
```

```
#include <sys/socket.h>
```

```
int socket(int family, int sort, int convention);
```

Here "family" will be AF_INET for IP correspondences, tradition will be zero, and sort will depend upon whether TCP or UDP is used. Two strategies wishing to pass on finished a framework make a connection each. These resemble two terminations of a pipe - however the bona fide pipe does not yet exist.

JFree Chart

JFreeChart is a free 100% Java diagram library that makes it basic for creators to indicate capable quality blueprints in their applications. JFreeChart's expansive rundown of capacities fuses:

A dependable and all around filed API, supporting a broad assortment of chart sorts;

A versatile arrangement that is definitely not hard to extend, and targets both server-side and client side applications;

Support for some yield sorts, including Swing parts, picture records

(checking PNG and JPEG), and vector plans archive gatherings (tallying PDF, EPS and SVG);

JFreeChart is "open source" or, more especially, free programming. It is scattered under the terms of the GNU Lesser General Public License (LGPL), which stipends use in prohibitive applications.

1. Guide Visualizations

Diagrams exhibiting values that relate to topographical locales. A couple of delineations include: (a) people thickness in each state of the United States,

(b) pay per capita for each country in Europe, (c) future in each country of the world. The errands in this wander include:

Sourcing openly redistributable vector plots for the nations of the world, states/regions specifically nations (USA specifically, yet additionally different territories);

Making a legitimate dataset interface (notwithstanding default utilization), a rendered, and organizing this with the current XYPlot class in JFreeChart; Testing, filing, testing some moreover, recording some more.

2. Time Series Chart Interactivity

Execute another (to JFreeChart) incorporate for instinctive time course of action plots - to demonstrate an alternate control that exhibits a little type of ALL the time plan data, with a sliding "view" rectangle that empowers you to pick the subset of the time game plan data to appear in the essential outline.

3. Dashboards

There is at introduce a lot of energy for dashboard appears. Influence a versatile dashboard to instrument that support a subset of JFreeChart diagram sorts (dials, pies,

thermometers, bars, and lines/time plan) that can be passed on successfully by methods for both Java Web Start and an applet.

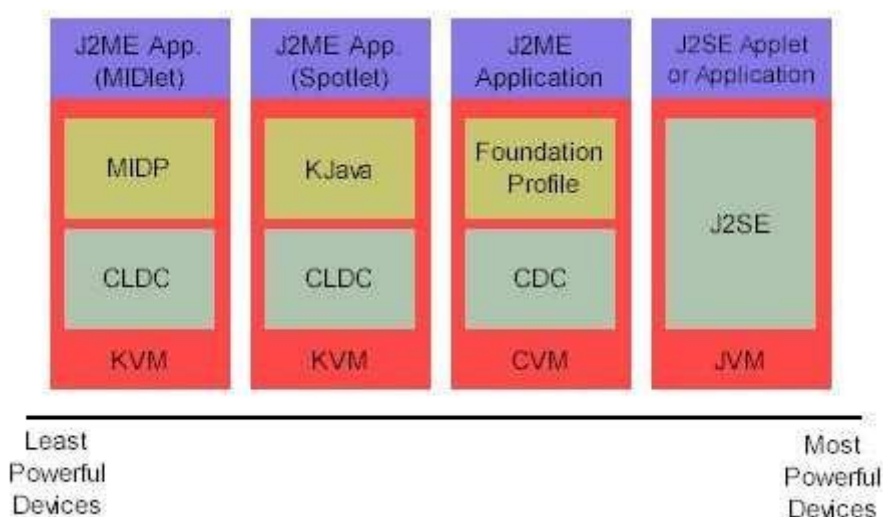
4. Property Editors

The property administrator instrument in JFreeChart just handles a little subset of the properties that can be set for graphs. Expand (or reimplement) this segment to give more essential end-customer control over the nearness of the outlines.

J2ME (Java 2 Micro form):-

Sun Microsystems describes J2ME as "an exceptionally upgraded Java run-time condition concentrating on a broad assortment of purchaser things, including pagers, PDAs, screen-phones, mechanized set-top boxes and auto course systems." Announced in June 1999 at the JavaOne Developer Conference, J2ME brings the cross-organize value of the Java vernacular to tinier devices, empowering adaptable remote contraptions to share applications. With J2ME, Sun has balanced the Java arrange for purchaser things that join or rely upon small figuring contraptions.

1. J2ME architecture



J2ME uses outlines and profiles to adjust the Java Runtime Environment (JRE). As an aggregate JRE, J2ME is contained a setup, which chooses the JVM used, and a profile, which describes the application by including space specific classes. The outline describes the fundamental run-time condition as a game plan of focus classes and a specific JVM that continue running on specific sorts of devices. We'll discuss plans in detail in the profile portrays the application; especially, it adds range specific classes to the J2ME setup to describe certain usages for devices. We'll cover profiles all around. The going with sensible depicts the association between the different virtual machines, outlines, and profiles. It also draws a parallel with the J2SE API and its Java virtual machine. While the J2SE virtual machine is generally insinuated as a JVM, the J2ME virtual machines,

KVM and CVM, are subsets of JVM. Both KVM and CVM can be thought of as a kind of Java virtual machine - it's as of late that they are contracted variations of the J2SE JVM and are specific to J2ME.

2. Making J2ME applications

Introduction:

Here, we will go over a few considerations you have to review while making applications for more diminutive contraptions. We'll look at the way the compiler is summoned while utilizing J2SE to orchestrate J2ME applications. At last, we'll investigate bundling and affiliation and the part precertification plays in this framework.

3. Outline considerations for little gadgets

Making applications for little contraptions suspects that you will review certain methods amidst the outline mastermind. It is best to deliberately plan an application for a little contraption before you start coding. Refreshing the code since you neglect to consider a large portion of the "gotchas" before working up the application can be a horrifying methodology. Here are some course of action systems to consider:

Keep it coordinate. Clear paltry highlights, conceivably making those highlights an other, optional application.

- * Smaller is better. This thought ought to be a "basic decision" for all makers. More minor applications utilize less memory on the gadget and require shorter establishment times. Consider bundling your Java applications as compacted Java Archive (shake) reports.

- * Minimize run-time memory utilize. To oblige the measure of memory utilized at run time, utilize scalar sorts set up of question sorts. Moreover, don't rely on the junk move. You ought to deal with the memory proficiently yourself by setting object references to invalid when you are done with them. Another approach to manage reduce run-time memory is to utilize unresponsive instantiation, basically assigning objects on an as-required present. Unmistakable strategies for diminishing all around and peak memory

use on little contraptions are to discharge assets rapidly, reuse request, and maintain a strategic distance from one of a kind cases,.

4. Setups audit

The setup describes the fundamental run-time condition as a course of action of focus classes and a specific JVM that continue running on specific sorts of contraptions. Starting at now, two courses of action exist for J2ME, in any case others may be described later on:

* Associated Limited Device Configuration (CLDC) is utilized particularly with the KVM for 16-bit or 32-bit contraptions with constrained measures of memory. This is the arrangement (and the virtual machine) utilized for growing little J2ME applications. Its size restrictions make CLDC in addition enrapturing and testing (from a change perspective) than CDC. CLDC is besides the setup that we will use for working up our depiction mechanical gathering application. A case of a little remote contraption running little applications is a Palm hand-held PC.

* Connected Device Configuration (CDC) is utilized with the C virtual machine (CVM) and is utilized for 32-bit structures requiring more than 2 MB of memory. An example of such a gadget is a Net TV box.

5. J2ME profiles

What is a J2ME profile?

As we said before in this instructional exercise, a profile portrays the kind of gadget strengthened. The Mobile Information Device Profile (MIDP), for instance, depicts classes for PDAs. It adds space particular classes to the J2ME game-plan to portray utilizes for all intents and purposes indistinguishable contraptions. Two profiles have been depicted for J2ME and rely upon CLDC: KJava and MIDP. Both KJava and MIDP are associated with CLDC and more little contraptions. Profiles are based over setups. Since profiles are particular to the measure of the gadget (measure of memory) on which an application runs, certain profiles are associated with specific strategies.

A skeleton profile whereupon you can make your own particular profile, the Foundation Profile, is open for CDC.

Profile 1: KJava

KJava is Sun's restrictive profile and contains the KJava API. The KJava profile is based over the CLDC design. The KJava virtual machine, KVM, perceives a near byte codes and class record arrange as the commendable J2SE virtual machine. KJava contains a Sun-particular API that keeps running on the Palm OS. The KJava API has an unfathomable game-plan in an indistinct path from the J2SE Abstract Windowing Toolkit (AWT). All things considered, in light of the way that it is not a standard J2ME bundle, its administer bundle is com.sun.kjava. We'll take in extra about the KJava API later in this instructional exercise when we build up some delineation applications.

Profile 2: MIDP

MIDP is intended for PDAs, for instance, cell phones and pagers. The MIDP, as KJava, depends on CLDC and gives a standard run-time condition that empowers new applications and organizations to be passed on continuously on end customer contraptions. MIDP is a run of the mill, business standard profile for mobile phones that is not dependent on a specific dealer. It is a whole and maintained foundation for versatile application

change. MIDP contains the going with groups, the underlying three of which are focus CLDC bundles, in addition to three MIDP-particular bundles.

- * java.lang
- * java.io
- * java.util
- * javax.microedition.io
- * javax.microedition.lcdui
- * javax.microedition.midlet
- * javax.microedition.rms

CHAPTER-5

SYSTEM DESIGN

5.1. INTRODUCTION

INPUT DESIGN

The data setup is the relationship between the data structure and the client. It fuses the making particular and system for information status and those techniques are basic to put exchange information in to a usable edge for arranging can be capable by assessing the PC to analyze information from a made or printed report or it can happen by having individuals entering the information unmistakably into the structure. The game plan of information concentrates on controlling the measure of information required, controlling the mistakes, keeping up a key division from delay, maintaining a strategic distance from additional methods and keeping the system basic. The information is orchestrated in such a path therefore, to the point that it gives security and convenience with holding the protection. Information Design considered the running with things:

What data should be given as data?

How the information ought to be dealt with or coded?

The exchange to arrange the working staff in giving information.

Techniques for getting ready information supports and sets out to take after when mess up happen.

Objectives

1. Data Design is the course toward changing over a client organized portrayal of the sense of duty regarding a PC based structure. This outline is fundamental to maintain a strategic distance from bungles in the information input process and show the right rushing toward the association for getting right data from the mechanized framework.

2. It is master by impacting easy to use screens for the information to fragment to oversee broad volume of information. The objective of organizing input is to impact information to

region less asking for and to be free from bumbles. The information range screen is delineated with the true objective that every single one of the information controls can be performed. It correspondingly gives record seeing work environments.

3. Right when the data is entered it will check for its authenticity. Data can be entered with the help of screens. Reasonable messages are given as when required with the objective that the customer won't be in maize of minute. In like manner the objective of data design is to influence an information to organize that is definitely not hard to take after

Yield DESIGN

A quality yield is one, which meets the necessities of the end client and presents the data plainly. In any framework possible results of preparing are gave to the clients and to other structure through yields. In yield design it is settled how the data is to be unstuck for affect require and in addition the printed variation yield. It is the most objective and direct source data to the client. Valuable and clever yield configuration redesigns the structure's relationship to help client principal specialist.

1. Organizing PC yield ought to continue in a managed, weakened totally considered way; the correct yield must be made while guaranteeing that each yield fragment is shaped so individuals will discover the structure can utilize effortlessly and viably. Right when examination graph PC yield, they should Identify the particular yield that is depended upon to meet the prerequisites.

Select strategies for showing data.

3. Impact archive, to report, or unmistakable affiliations that contain data made by the structure.

The yield sort of a data framework ought to fulfill no short of what one of the running with targets.

Pass on data about past exercises, current status or projections of the

Future.

Banner basic events, openings, issues, or takes note.

Trigger an action.

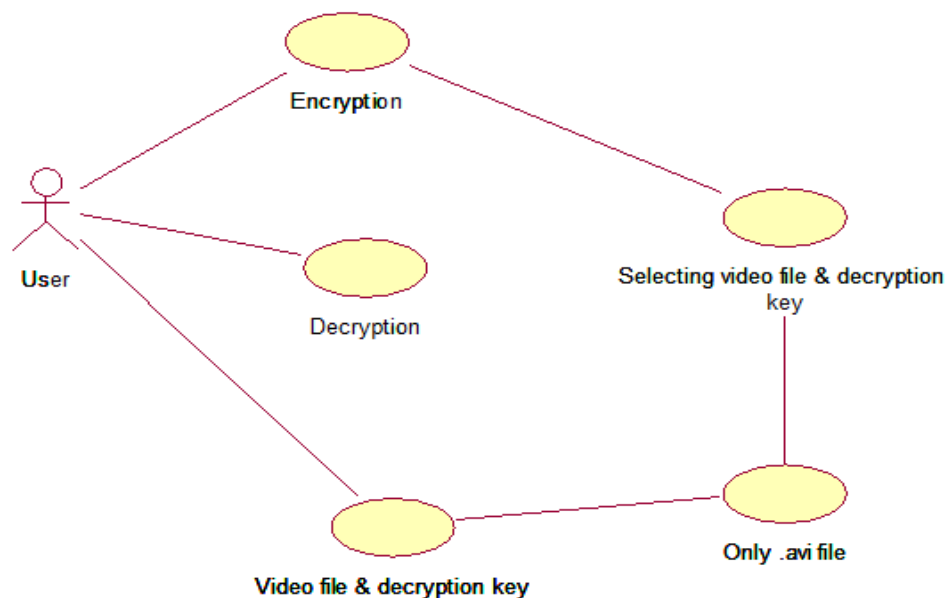
Confirm an action.

5.1.1 SYSTEM ARCHITECTURE

Use Case Diagram / Sequence Diagram/ Collaboration Diagram / Class Diagram:

5.2 Use Case Diagram:

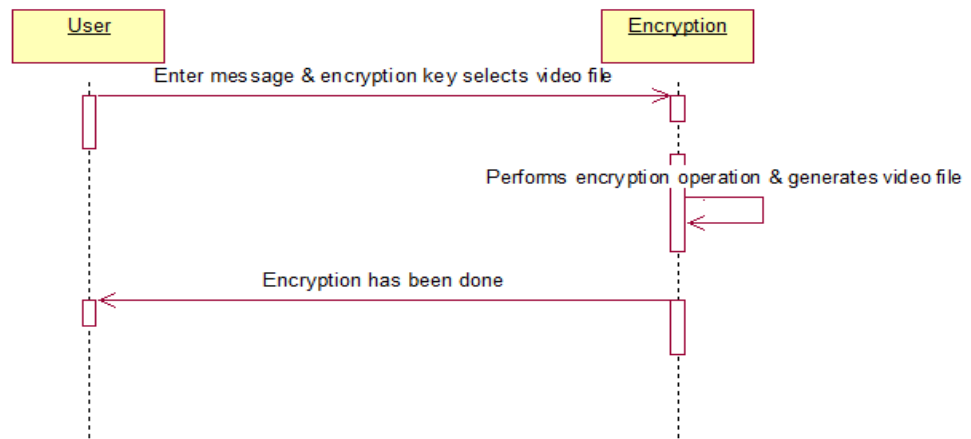
Utilize case outline demonstrates an arrangement of utilization case and performing artists (a unique sort of class) and their connections. Utilize case outlines address the static utilize case perspective of a framework. These outlines are particularly imperative in sorting out and demonstrating the conduct of a framework. (Admin)



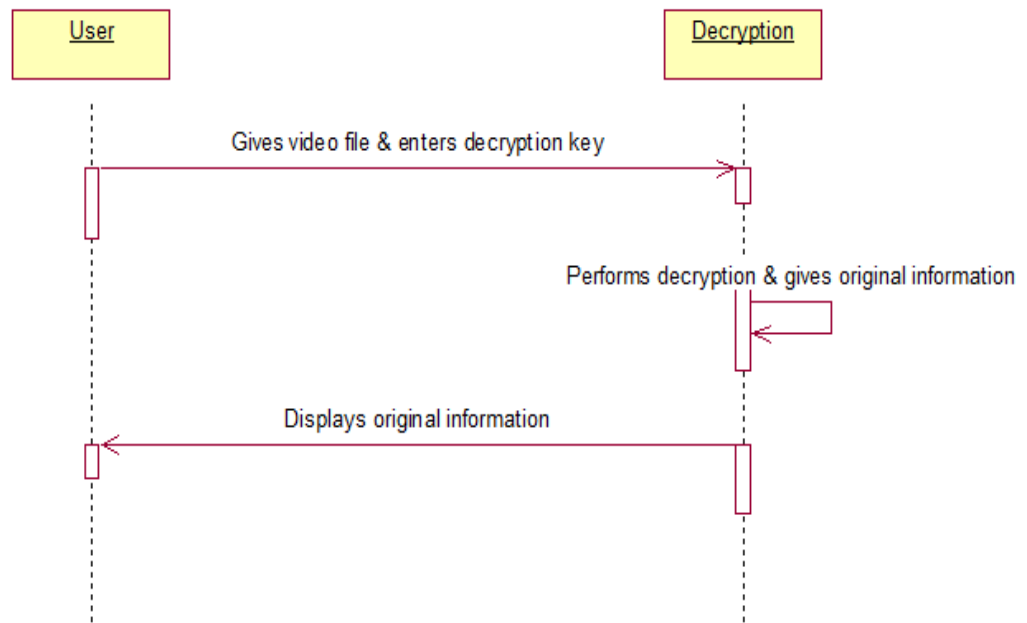
5.2 Use Case Diagram

5.3 Sequence Diagram:

Sequence diagram is an interaction diagram that emphasizes the time-ordering of messages



5.3.1 Sequence Diagram

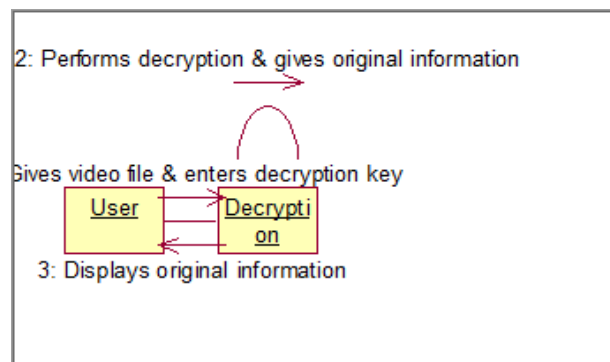


5.3.2 Sequence Diagram

5.4 Collaboration diagrams

Collaboration Diagram Show as the chain of responsibilities.

5.4.1 Collaboration Diagram

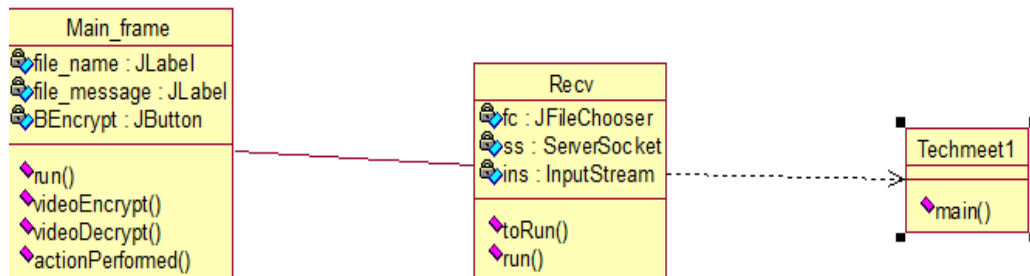


5.4.2 Collaboration Diagram

5.5 Class Diagram

Class graph demonstrates an arrangement of classes, interfaces, and joint efforts and their connections. Class charts address the static outline perspective of a framework. Class

charts that incorporate dynamic classes address the static procedure perspective of a framework.



5.5 Class Diagram

CHAPTER-6

SAMPLE CODE

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;
import java.io.File;
import java.util.Hashtable;
public class EmbedFileGUI extends JFrame implements ActionListener,
ItemListener{
    BackEndHandler client;
    private JLabel lblMaster, lblOutput, lblData, lblMasterSize, lblOutputSize,
    lblMessage;;
    private JLabel lblMasterFileSize, lblOutputFileSize, lblStatus;
    private JLabel lblDataSize, lblDataFileSize;
    private JLabel lblCompression, lblPassword;
    private JCheckBox chkCompress, chkEncrypt;
    private JSlider sliderCompression;\
    private JPasswordField txtPassword;
    private JTextField txtMasterFile, txtOutputFile, txtDataFile;
    private JScrollPane scrollPane;
    private JButton btnGo, btnHelp, btnCancel;
    private JButton btnChangeMasterFile, btnChangeOutputFile, btnChangeDataFile;
    public EmbedFileGUI(BackEndHandler client)
    {
        super("Embedding file - Steganograph");
        setDefaultCloseOperation(DISPOSE_ON_CLOSE);        this.client= client;
```

```

Font arialFont= new Font("Arial", Font.PLAIN, 11);
lblMaster= new JLabel("Master file");
lblOutput= new JLabel("Output file");
lblData=      new JLabel("Data file");
lblMasterSize= new JLabel("Size: ");
lblOutputSize= new JLabel("Size: ");
lblDataSize=  new JLabel("Size: ");
txtMasterFile= new MyJTextField(client.getMasterFile().getName(),
13, Color.blue, Color.lightGray);
txtMasterFile.setEditable(false);
lblMasterFileSize=      new      MyJLabel(""+
client.getMasterFile().length()/1024+ " Kb", arialFont, Color.red, Color.gray);
txtOutputFile= new MyJTextField(client.getOutputFile().getName(), 13, Color.blue,
Color.lightGray);
txtOutputFile.setEditable(false);
lblOutputFileSize=  new  MyJLabel(lblMasterFileSize.getText(),
arialFont, Color.red, Color.gray);
txtDataFile=  new  MyJTextField(client.getDataFile().getName(), 13, Color.blue,
Color.lightGray);
txtDataFile.setEditable(false);
lblDataFileSize=      new      MyJLabel(""+
client.getDataFile().length()/1024+ " Kb", arialFont, Color.red, Color.gray);
btnChangeMasterFile= new JButton("Change");
btnChangeOutputFile= new JButton("Change");
btnChangeDataFile=  new JButton("Change");
btnGo= new JButton("  Go  ");
btnHelp= new JButton("  Help  ");
btnCancel= new JButton("  Close  ");
lblCompression= new MyJLabel("Compression level", arialFont, Color.black,
Color.lightGray);
lblPassword=  new  MyJLabel("Password (Minimum 8 chars)",
arialFont, Color.black, Color.lightGray);
chkCompress= new JCheckBox("Compress");

```

```

chkEncrypt= new JCheckBox("Encrypt");
        sliderCompression= new JSlider(0, 9, 5);
sliderCompression.setForeground(Color.blue);
sliderCompression.setPaintTicks(true);
sliderCompression.setPaintLabels(true);
sliderCompression.setSnapToTicks(true);
sliderCompression.setMajorTickSpacing(1);
Hashtable h= new Hashtable();
h.put(new Integer(0), new JLabel("0".toString(), JLabel.CENTER));
h.put(new Integer(5), new JLabel("5".toString(), JLabel.CENTER));
h.put(new Integer(9), new JLabel("9".toString(), JLabel.CENTER));
sliderCompression.setLabelTable(h);

        txtPassword= new JPasswordField(9);
chkCompress.addItemListener(this);
chkEncrypt.addItemListener(this);
lblCompression.setEnabled(false);
sliderCompression.setEnabled(false);
lblPassword.setEnabled(false);
txtPassword.setEnabled(false);

        // Setup panel file1
JPanel file1= new JPanel();
new BoxLayout(file1, BoxLayout.X_AXIS);
file1.add(lblMaster);
file1.add(txtMasterFile);
file1.add(lblMasterSize);
file1.add(lblMasterFileSize);
file1.add(btnChangeMasterFile);

// Setup panel file2
JPanel file2= new JPanel();
new BoxLayout(file2, BoxLayout.X_AXIS);
file2.add(lblOutput);
file2.add(txtOutputFile);
file2.add(lblOutputSize);
file2.add(lblOutputFileSize);

```



```

file2.add(btnChangeOutputFile);
                                // Setup panel file3
JPanel file3= new JPanel();
new BoxLayout(file3, BoxLayout.X_AXIS);
file3.add(lblData);
file3.add(txtDataFile);
file3.add(lblDataSize);
file3.add(lblDataFileSize);
file3.add(btnChangeDataFile);

JPanel panelFiles= new JPanel();
GridBagLayout gbl= new GridBagLayout();
GridBagConstraints gbc= new GridBagConstraints();
panelFiles.setLayout(gbl);
JLabel lblFiller;
gbc.anchor= gbc.WEST;      gbc.gridx=1; gbc.gridy= 1;
gbl.setConstraints(file1, gbc);
panelFiles.add(file1);
lblFiller= new JLabel(" ");
gbc.gridy= 2;      gbl.setConstraints(lblFiller, gbc);
panelFiles.add(lblFiller);
gbc.gridy= 3;      gbl.setConstraints(file2, gbc);
panelFiles.add(file2);
lblFiller= new JLabel(" ");
gbc.gridy= 4;      gbl.setConstraints(lblFiller, gbc);
panelFiles.add(lblFiller);
gbc.gridy= 5;      gbl.setConstraints(file3, gbc);
panelFiles.add(file3);
panelFiles= UtilityOperations.createBorderedPanel(panelFiles, "Files",
19, 3);

// Setup features panel 1
JPanel panelFeatures1a= new JPanel();
new BoxLayout(panelFeatures1a, BoxLayout.X_AXIS);
panelFeatures1a.add(chkCompress);

```

```

        panelFeatures1a.add(lblCompression);
        JPanel panelFeatures1b= new JPanel();
        new BoxLayout(panelFeatures1b, BoxLayout.X_AXIS);
        panelFeatures1b.add(new MyJLabel("Low", arialFont, Color.blue,
Color.lightGray));
        panelFeatures1b.add(sliderCompression);
        panelFeatures1b.add(new MyJLabel("High", arialFont, Color.blue,
Color.lightGray));

        JPanel panelFeatures1= new JPanel();
        gbl= new GridBagLayout();
        panelFeatures1.setLayout(gbl);
        gbc.gridx= 1; gbc.gridy= 1; gbl.setConstraints(panelFeatures1a, gbc);
        panelFeatures1.add(panelFeatures1a);
        gbc.gridy= 2; gbl.setConstraints(panelFeatures1b, gbc);
        panelFeatures1.add(panelFeatures1b);
        panelFeatures1=
UtilityOperations.createBorderedPanel(panelFeatures1, "Compression", 10, 3);

        // Setup features panel 2
        JPanel panelFeatures2= new JPanel();
        new BoxLayout(panelFeatures2, BoxLayout.X_AXIS);
        panelFeatures2.add(chkEncrypt);
        panelFeatures2.add(lblPassword);
        panelFeatures2.add(txtPassword);
        panelFeatures2=
UtilityOperations.createBorderedPanel(panelFeatures2, "Encryption", 3, 3);

        JPanel panelFeatures= new JPanel();
        gbl= new GridBagLayout();
        panelFeatures.setLayout(gbl);
        gbc.anchor= gbc.WEST;      gbc.gridx=1; gbc.gridy= 1;
        gbl.setConstraints(panelFeatures1, gbc);
        panelFeatures.add(panelFeatures1);
        lblFiller= new JLabel(" ");
        gbc.gridy= 2;      gbl.setConstraints(lblFiller, gbc);
        panelFeatures.add(lblFiller);

```

```

gbc.gridy= 3;          gbl.setConstraints(panelFeatures2, gbc);
panelFeatures.add(panelFeatures2);
                // Setup the buttons panel
JPanel panelButtons= new JPanel();
gbl= new GridBagLayout();
panelButtons.setLayout(gbl);
gbc.anchor= gbc.WEST;   gbc.gridx=1; gbc.gridy= 1;
gbl.setConstraints(btnGo, gbc);
panelButtons.add(btnGo);
lblFiller= new JLabel(" ");
gbc.gridy= 2;          gbl.setConstraints(lblFiller, gbc);
panelButtons.add(lblFiller);
gbc.gridy= 3;          gbl.setConstraints(btnHelp, gbc);
panelButtons.add(btnHelp);
lblFiller= new JLabel(" ");
gbc.gridy= 4;          gbl.setConstraints(lblFiller, gbc);
panelButtons.add(lblFiller);
gbc.gridy= 5;          gbl.setConstraints(btnCancel, gbc);
panelButtons.add(btnCancel);
                JPanel mainPanel= new JPanel();
new BoxLayout(mainPanel, BoxLayout.Y_AXIS);
mainPanel.add(panelFiles);
mainPanel.add(new JLabel(" "));
mainPanel.add(panelFeatures);
mainPanel.add(new JLabel(" "));
mainPanel.add(panelButtons);
mainPanel= UtilityOperations.createBorderedPanel(mainPanel, 3, 3);
setContentPane(mainPanel);
                btnChangeMasterFile.addActionListener(this);
btnChangeOutputFile.addActionListener(this);
btnChangeDataFile.addActionListener(this);
btnHelp.addActionListener(this);
btnGo.addActionListener(this);
btnCancel.addActionListener(this);

```

```

        Dimension d= Toolkit.getDefaultToolkit().getScreenSize();
        int width= (int)(0.7* d.width);
        int height= (int)(0.85* d.height);
        setSize(width, height);
        setLocation((d.width-        width)/2,        (d.height-        height)/2
//setResizable(false);
        setVisible(true);
    }

    public void actionPerformed(ActionEvent e)
    {
        Object source= e.getSource();
        if(source== btnChangeMasterFile)
        {
            client.chooseMasterFile();
            txtMasterFile.setText(client.getMasterFile().getName());
            lblMasterFileSize.setText(""+
client.getMasterFile().length()/1024+ "Kb");
            lblOutputFileSize.setText(lblMasterFileSize.getText());
        }

        if(source== btnChangeOutputFile)
        {
            client.chooseOutputFile();
            txtOutputFile.setText(client.getOutputFile().getName());
        }

        if(source== btnChangeDataFile)
        {
            client.chooseDataFile();
            txtDataFile.setText(client.getDataFile().getName());
            lblDataFileSize.setText(""+ client.getDataFile().length()/1024+
"Kb");
        }

        if(source== btnHelp)
            Steganograph.showHelpDialog();
        if(source== btnCancel)

```

```

        dispose();

        if(source== btnGo)
        {
            int compression= sliderCompression.getValue();
            String password= null;

            if(chkEncrypt.isSelected())
            {
                password= new String(txtPassword.getPassword());
                if(password.length()<8)
                {
                    JOptionPane.showMessageDialog(this,
"Password needs to be a minimum of 8 Characters!", "Invalid password!",
JOptionPane.WARNING_MESSAGE);

                    return;
                }
            }

            if(client.getOutputFile().exists())
            {
                int result2= JOptionPane.showConfirmDialog(null,
"File "+ client.getOutputFile().getName()+ " already exists!\nWould you like to
OVERWRITE it?", "File already exists!", JOptionPane.YES_NO_OPTION);
                if(!(result2== JOptionPane.YES_OPTION))
                if(!client.chooseOutputFile())
                    return;
            }

            if(Steganograph.embedFile(client.getMasterFile(), client.getOutputFile(),
client.getDataFile(), compression, password))
                JOptionPane.showMessageDialog(this,
Steganograph.getMessage(), "Operation Successful",
JOptionPane.INFORMATION_MESSAGE);
            else

```

```

        JOptionPane.showMessageDialog(this,
Steganograph.getMessage(),          "Operation Unsuccessful",
JOptionPane.ERROR_MESSAGE);
    }
}
public void itemStateChanged(ItemEvent e)
{
    if(e.getSource()== chkCompress)
    {
        if(chkCompress.isSelected())
        {
            lblCompression.setEnabled(true);
            sliderCompression.setEnabled(true);
        }
        else
        {
            lblCompression.setEnabled(false);
            sliderCompression.setEnabled(false);
        }
    }
    else
    {
        if(chkEncrypt.isSelected())
        {
            lblPassword.setEnabled(true);
            txtPassword.setEnabled(true);
        }
        else
        {
            lblPassword.setEnabled(false);
            txtPassword.setEnabled(false);
        }
    }
}
}

```

```

}
import javax.swing.*;
import javax.swing.border.TitledBorder;
import java.awt.*;
import java.awt.event.*;
public class MainClient extends WindowAdapter implements ActionListener
{
    private JFrame      mainFrame;
    private JMenuBar    menuBar;
    private JMenu       menuFile,    menuEdit,    menuView,    menuHelp,
menuLookAndFeel;
    private JMenuItem   mnuExit,      mnuEmbedMessage,    mnuEmbedFile,
mnuHelp, mnuAbout;
    private JMenuItem   mnuRetrieveMessage,                mnuRetrieveFile,
mnuModifyMaster;
    private JRadioButtonMenuItem    mnuTonicFeel,          mnuMetalFeel,
mnuMotifFeel, mnuWindowsFeel;
    private ButtonGroup lookAndFeelButtonGroup;
private JPanel mainPanel, panelAbout, panelButtons;
    private JLabel lblLogo;
    private JLabel lblFiller[], lblName, lblEmail, lblPhone;
    private GridBagLayout gbl;
    private GridBagConstraints gbc;
    private MyJButton btnEmbedFile, btnRetrieveFile, btnEmbedMessage,
btnRetrieveMessage;
    private MyJButton btnHelp, btnAbout;
    private BackEndHandler back;
    private MainClient()
    {
        mainFrame= new JFrame("venkat");
        mainFrame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        mainFrame.addWindowListener(this);
        // Setup the menu bar
        mnuExit= new JMenuItem("Exit", 1, 'x');

```

```

mnuEmbedMessage= new MyJMenuItem("Embed Message", 6, 'm');
mnuEmbedFile= new MyJMenuItem("Embed File", 7, 'i');
mnuRetrieveMessage= new MyJMenuItem("Retrieve Message", 0, 'r');
mnuRetrieveFile= new MyJMenuItem("Retrieve File", 2, 't');
mnuModifyMaster= new MyJMenuItem("Modify Master file settings", 2, 'd');
mnuHelp= new MyJMenuItem("Help", 0, 'h');
mnuAbout= new MyJMenuItem("About", 0, 'a');
mnuTonicFeel= new MyJRadioButtonMenuItem("Plastic XP", 8, 'x');
mnuMetalFeel= new MyJRadioButtonMenuItem("Metal", 0, 'm');
mnuMotifFeel= new MyJRadioButtonMenuItem("Motif", 2, 't');
mnuWindowsFeel= new MyJRadioButtonMenuItem("Windows", 0,
'w');

// Add item listener for Look and feel menu items
RadioListener radioListener= new RadioListener();
mnuTonicFeel.addItemListener(radioListener);
mnuMetalFeel.addItemListener(radioListener);
mnuMotifFeel.addItemListener(radioListener);
mnuWindowsFeel.addItemListener(radioListener);
mnuTonicFeel.setSelected(true);
lookAndFeelButtonGroup= new ButtonGroup();
lookAndFeelButtonGroup.add(mnuTonicFeel);
lookAndFeelButtonGroup.add(mnuMetalFeel);
lookAndFeelButtonGroup.add(mnuMotifFeel);
lookAndFeelButtonGroup.add(mnuWindowsFeel);
// Add action listeners for other menu items
mnuEmbedMessage.addActionListener(this);
mnuEmbedFile.addActionListener(this);
mnuRetrieveMessage.addActionListener(this);
mnuRetrieveFile.addActionListener(this);
mnuModifyMaster.addActionListener(this);
mnuExit.addActionListener(this);
mnuHelp.addActionListener(this);
mnuAbout.addActionListener(this);
menuFile= new MyJMenu("File", 0, 'f');

```



```

menuFile.add(mnuEmbedMessage);
menuFile.add(mnuEmbedFile);
menuFile.add(mnuRetrieveMessage);
menuFile.add(mnuRetrieveFile);
menuFile.add(mnuExit);
menuEdit= new JMenu("Edit");
menuEdit.add(mnuModifyMaster);
menuLookAndFeel= new MyJMenu("Look and Feel...", 0, 'l');
menuLookAndFeel.add(mnuTonicFeel);
menuLookAndFeel.add(mnuMetalFeel);
menuLookAndFeel.add(mnuMotifFeel);
menuLookAndFeel.add(mnuWindowsFeel);
menuView= new MyJMenu("View", 0, 'v');
menuView.add(menuLookAndFeel);
menuHelp= new MyJMenu("Help", 0, 'h');
menuHelp.add(mnuHelp);
//menuHelp.add(mnuAbout);
menuBar= new JMenuBar();
menuBar.add(menuFile);
//menuBar.add(menuEdit);
//menuBar.add(menuView);
menuBar.add(menuHelp);
mainFrame.setJMenuBar(menuBar);
mainPanel= new JPanel();
panelAbout= new JPanel();
panelButtons= new JPanel();
// Create filler labels
lblFiller= new JLabel[4];
for(int i=0; i<4; i++)
    lblFiller[i]= new JLabel(" ");
// Prepare About panel
gbl= new GridBagLayout();
gbc= new GridBagConstraints();
panelAbout.setLayout(gbl);

```

```

panelAbout.setBackground(Color.white);
Color myColor= new Color(50, 153, 237);
Font arialFont= new Font("Arial", Font.PLAIN, 14);
Font myFont= new Font("Monotype Corsiva", Font.PLAIN, 18);
// Prepare the Buttons panel
panelButtons.setBackground(Color.white);
gbl= new GridBagLayout();
panelButtons.setLayout(gbl);
panelButtons.setBorder(new TitledBorder("Supported operations"));
//lblLogo=
new JLabel(new
ImageIcon("D://Nidheesh//jins//Images//nnn.jpeg"));
lblLogo= new JLabel(new ImageIcon("./Images/Logo.png"));
btnEmbedMessage=
new
MyJButton("./Images/EmbedMessage.bmp", "./Images/EmbedMessageHover.bmp");
btnEmbedFile=
new MyJButton("./Images/EmbedFile.bmp",
"./Images/EmbedFileHover.bmp");
btnRetrieveMessage=
new
MyJButton("./Images/RetrieveMessage.bmp",
"./Images/RetrieveMessageHover.bmp");
btnRetrieveFile=
new MyJButton("./Images/RetrieveFile.bmp",
"./Images/RetrieveFileHover.bmp");
btnHelp=
new MyJButton("./Images/Help.bmp",
"./Images/HelpHover.bmp");
btnAbout=
new MyJButton("./Images/About.bmp",
"./Images/AboutHover.bmp");

// Add action listeners for the buttons
btnEmbedMessage.addActionListener(this);
btnEmbedFile.addActionListener(this);
btnRetrieveMessage.addActionListener(this);
btnRetrieveFile.addActionListener(this);
btnHelp.addActionListener(this);
btnAbout.addActionListener(this);

// Add filler for rows 1 and 2
gbc.weightx= 4;      gbc.weighty= 2;      gbc.fill= gbc.BOTH;

```

```

gbc.gridx= 6; gbc.gridy= 1; gbl.setConstraints(lblFiller[0], gbc);
panelButtons.add(lblFiller[0]);
gbc.weightx= 1;      gbc.weighty= 1;      gbc.fill= gbc.NONE;
gbc.gridx= 3; gbc.gridy= 3; gbl.setConstraints(btnHelp, gbc);
//panelButtons.add(btnHelp);
gbc.gridx= 5; gbl.setConstraints(btnAbout, gbc);
//panelButtons.add(btnAbout);
// Add filler for rows 4 and 5
gbc.fill = gbc.BOTH;
gbc.gridx= 1; gbc.weighty= 2;      gbc.gridy= 4;
gbl.setConstraints(lblFiller[1], gbc);
panelButtons.add(lblFiller[1]);
gbc.fill= gbc.NONE;
gbc.gridx= 2; gbc.weighty= 1;      gbc.gridy= 6;
gbl.setConstraints(btnEmbedMessage, gbc);
panelButtons.add(btnEmbedMessage);
gbc.gridx= 4; gbl.setConstraints(btnRetrieveMessage, gbc);
panelButtons.add(btnRetrieveMessage);
// Add filler for row 7 and 8
gbc.fill = gbc.BOTH;
gbc.gridx= 6; gbc.weighty= 2;      gbc.gridy= 7;
gbl.setConstraints(lblFiller[2], gbc);
panelButtons.add(lblFiller[2]);
gbc.fill= gbc.NONE;
gbc.gridx= 3; gbc.weighty= 1;      gbc.gridy= 9;
gbl.setConstraints(btnEmbedFile, gbc);
panelButtons.add(btnEmbedFile);
gbc.gridx= 5; gbl.setConstraints(btnRetrieveFile, gbc);
panelButtons.add(btnRetrieveFile);

// Add the lblLogo and two Panels to the mainPanel
gbl= new GridBagLayout();
mainPanel.setLayout(gbl);
mainPanel.setBackground(Color.white);

```

```

        gbc.anchor= gbc.CENTER;
        gbc.gridx= 1; gbc.gridy= 1; gbc.weighty= 2;        gbc.fill=
gbc.VERTICAL;
        gbl.setConstraints(lblLogo, gbc);
        mainPanel.add(lblLogo);
        gbc.gridy= 3; gbc.weighty= 2;
        gbl.setConstraints(panelAbout, gbc);
        mainPanel.add(panelAbout);
        gbc.gridy= 5; gbc.weighty= 1;
        gbl.setConstraints(panelButtons, gbc);
        mainPanel.add(panelButtons);
        gbc.gridy= 6; gbc.weighty= 2;
        gbl.setConstraints(lblFiller[3], gbc);
        mainPanel.add(lblFiller[3]);
        JPanel tempPanel= (JPanel) mainFrame.getContentPane();
        tempPanel.add(mainPanel, BorderLayout.CENTER);
        tempPanel.add(new MyJLabel(" ", Color.black, Color.darkGray),
BorderLayout.SOUTH);

        Dimension d= Toolkit.getDefaultToolkit().getScreenSize();
        mainFrame.setSize(d.width, (int) (d.height-(d.height*.03)));
        mainFrame.setResizable(false);
        mainFrame.setVisible(true);
    }

    // Listener methods
    public void actionPerformed(ActionEvent e)
    {
        Object source= e.getSource();
        // Embed message operation
        if(source== mnuEmbedMessage || source== btnEmbedMessage)
        {
            back= new BackEndHandler(this,
BackEndHandler.EMBED_MESSAGE);
            back.start();
        }
    }

```

```

    }
    // Retrieve message operation
    if(source== mnuRetrieveMessage || source== btnRetrieveMessage)
    {
        back= new BackEndHandler(this,
BackEndHandler.RETRIEVE_MESSAGE);
        back.start();
    }
    // Embed file operation
    if(source== mnuEmbedFile || source== btnEmbedFile )
    {
        back= new BackEndHandler(this,
BackEndHandler.EMBED_FILE);
        back.start();
    }
    // Retrieve file operation
    if(source== mnuRetrieveFile || source== btnRetrieveFile )
    {
        back= new BackEndHandler(this,
BackEndHandler.RETRIEVE_FILE);
        back.start();
    }
    // Modify Master file operation
    if(source== mnuModifyMaster)
    {
        back= new BackEndHandler(this,
BackEndHandler.EDIT_MASTER);
        back.start();
    }
    if(source== mnuHelp || source==btnHelp)
        Steganograph.showHelpDialog();
    if(source== mnuAbout || source== btnAbout)
        Steganograph.showAboutDialog();
    if(source== mnuExit)

```

```

        {
            int result= JOptionPane.showConfirmDialog(mainFrame, "Are
you sure that you want to close .", "Confirm Exit", JOptionPane.YES_NO_OPTION);
            if(result== JOptionPane.YES_OPTION)
            {
                System.exit(0);
            }
        }
    }
    // Class for lissoning to Look and feel radio menu events
    private class RadioListener implements ItemListener
    {
        public void itemStateChanged(ItemEvent e)
        {
            JMenuItem item= (JMenuItem) e.getSource();
            try
            {
                if(item== mnuTonicFeel && mnuTonicFeel.isSelected())

                UIManager.setLookAndFeel("com.jgoodies.looks.plastic.PlasticXPLookAndF
eel");

                if(item== mnuMetalFeel && mnuMetalFeel.isSelected())

                UIManager.setLookAndFeel("javax.swing.plaf.metal.MetalLookAndFeel");

                if(item== mnuMotifFeel && mnuMotifFeel.isSelected())

                UIManager.setLookAndFeel("com.sun.java.swing.plaf.motif.MotifLookAndF
eel");

                if(item== mnuWindowsFeel &&
mnuWindowsFeel.isSelected())
                UIManager.setLookAndFeel("com.sun.java.swing.plaf.windows.WindowsLookAndF
eel");

                SwingUtilities.updateComponentTreeUI(mainFrame);
                Steganograph.updateUserInterface();
            }
        }
    }
}

```

```
        }
        catch(Exception ex)
        {
        }
    }
}

// Main method
public static void main(String args[])
{
    new MainClient();
}
}
```

CHAPTER-7

TESTING

The clarification behind testing is to find messes up. Testing is the way toward trying to find each possible blame or deficiency in a work thing. It gives an approach to manage check the handiness of sections, sub-social events, congregations and what's more a completed the process of thing It is the way toward dealing with programming with the purpose of guaranteeing that the Software structure fulfills its necessities and client needs and does not flop in an inadmissible way. There are differing sorts of test. Each test sort watches out for a particular testing need.

CLASSIFICATION OF TESTS

7.1 UNIT TESTING

Unit testing incorporates the layout of investigations that support that within program justification is working fittingly, and that program inputs convey generous yields. All decision branches and inside code stream should be endorsed. It is the attempting of individual programming units of the application .it is done after the satisfaction of an individual unit before compromise. This is a helper testing, that relies upon learning of its advancement and is prominent. Unit tests perform basic tests at part level and test a specific business process, application, and also structure game plan. Unit tests ensure that each stand-out method for a business strategy performs decisively to the detailed conclusions and contains unmistakably portrayed sources of info and expected results.

7.2 INTEGRATION TESTING

Joining tests are intended to test incorporated programming parts to decide whether they really keep running as one program. Testing is occasion driven and is more worried about the fundamental result of screens or fields. Coordination tests exhibit that in spite of the fact that the parts were independently fulfillment, as appeared by effectively unit testing, the mix of segments is right and reliable. Joining testing is particularly gone for uncovering the issues that emerge from the blend of segments.

FUNCTIONAL TEST

Utilitarian tests give orderly exhibits that capacities tried are accessible as determined by the business and specialized necessities, framework documentation, and client manuals.

Practical testing is focused on the accompanying things:

Legitimate Input : recognized classes of substantial information must be acknowledged.

Invalid Input : recognized classes of invalid information must be rejected. Capacities : recognized capacities must be worked out.

Yield : distinguished classes of use yields must be worked out. Frameworks/Procedures : interfacing frameworks or strategies must be summoned.

Affiliation and arranging of reasonable tests is based on essentials, key limits, or exceptional trials. In addition, exact extension identifying with perceive Business process streams; data fields, predefined structures, and dynamic methods must be considered for testing. Before valuable testing is done, additional tests are recognized and the intense estimation of current tests is settled.

SYSTEM TEST

Structure testing ensures that the entire joined programming system meets requirements. It tests a setup to ensure known and obvious results. An instance of system testing is the outline orchestrated structure compromise test. Structure testing relies upon process portrayals and streams, underscoring pre-driven process associations and consolidation centers.

WHITE BOX TESTING

White Box Testing is an attempting in which in which the item analyzer thinks about the internal workings, structure and tongue of the item, or if nothing else its inspiration. It is reason. It is used to test domains that can't be come to from a disclosure level.

BLACK BOX TESTING

Divulgence Testing will be endeavoring the thing without any information of within workings, structure or vernacular of the module being endeavored. Exposure tests, as most one of a kind sorts of tests, must be shaped from a persuading source report, for example, particular or necessities record, for example, confirmation or basics document. It is an endeavoring in which the thing under test is overseen, as a revelation you can't "see" into it. The test gives wellsprings of data and reacts to yields without considering how the thing limits.

UNIT TESTING:

Unit testing is regularly driven as a noteworthy part of a joined code and unit trial of the item lifecycle, despite the way that it is typical for coding and unit testing to be coordinated as two specific stages

TEST STRATEGY AND APPROACH

Field testing will be performed physically and utilitarian tests will be composed in detail.

TEST OBJECTIVES

All field passages must work legitimately.

Pages must be actuated from the recognized connection.

The passage screen, messages and reactions must not be postponed.

HIGHLIGHTS TO BE TESTED

Confirm that the passages are of the right arrangement

No copy passages ought to be permitted

All connections should take the client to the right page.

COMBINED TESTING

Programming blend testing is the incremental joining testing of no less than two facilitated programming sections on a lone stage to convey frustrations caused by interface relinquishes.

The errand of the coordination test is to watch that fragments or programming applications, e.g. parts in an item structure or – one phase up – programming applications at the association level – work together without botch.

Test results: All the trials indicated above passed successfully. No defects experienced.

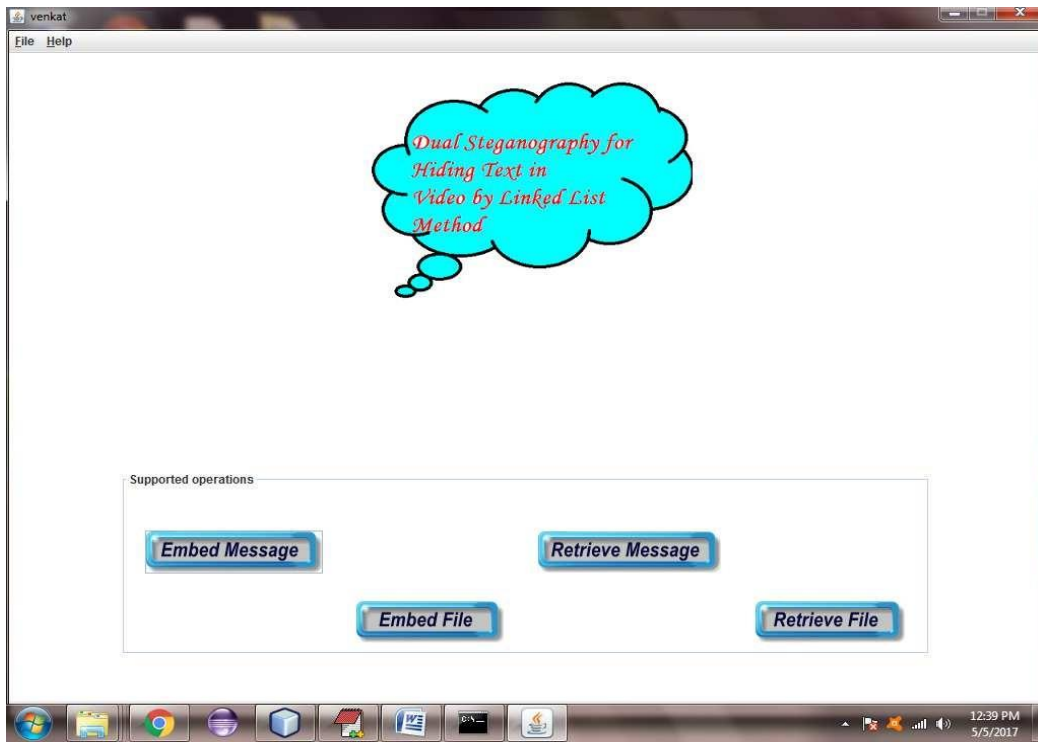
7.3 ACCEPTANCE TESTING

Customer Acceptance Testing is a fundamental time of any endeavor and requires basic enthusiasm by the end customer. It also ensures that the system meets the valuable requirements.

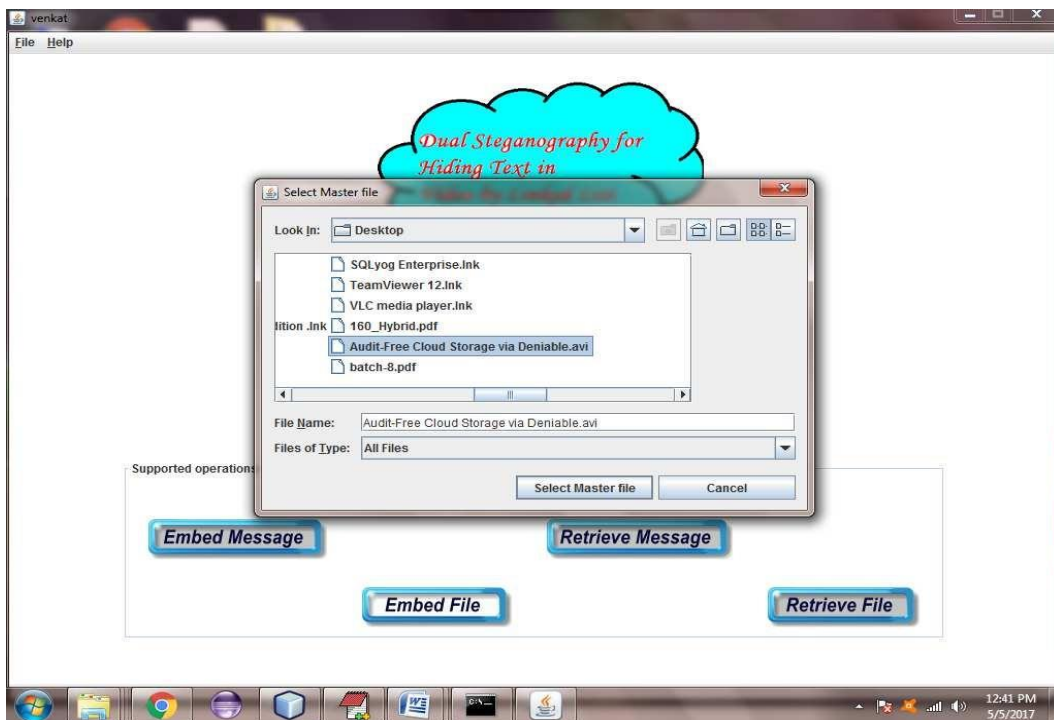
Test results: All the examinations said above passed adequately. No disfigurements experienced.

CHAPTER-8

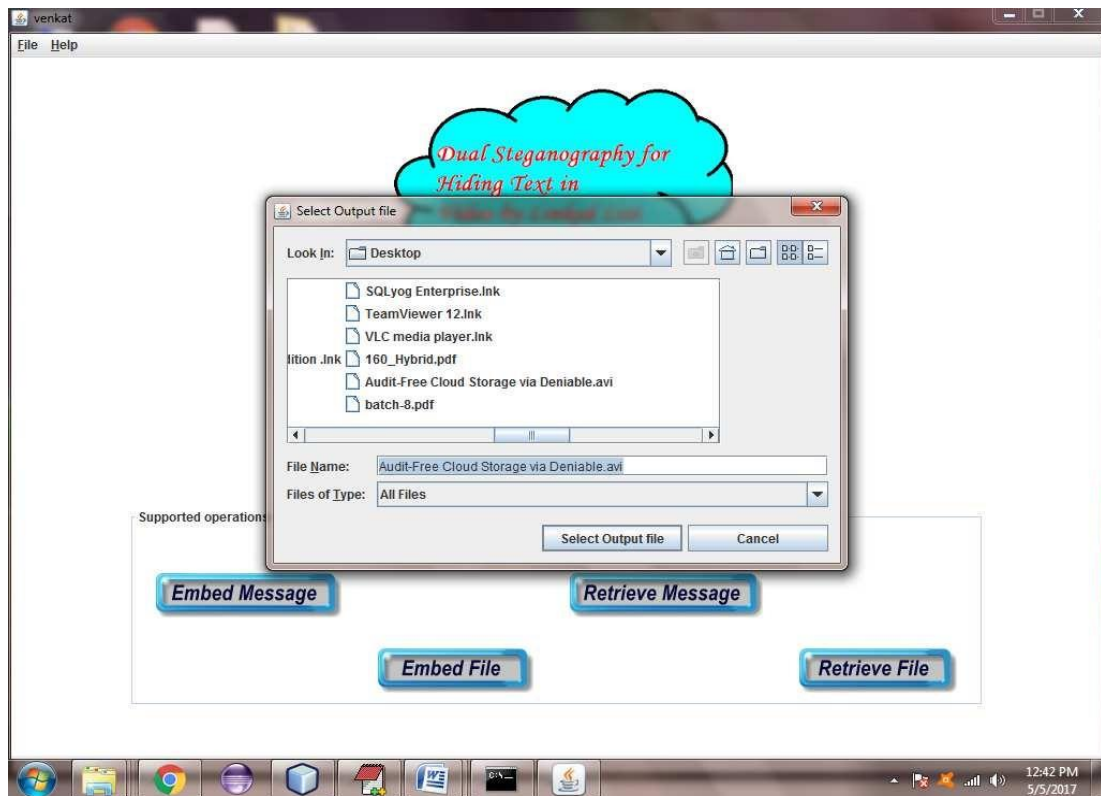
SCREEN SHOTS



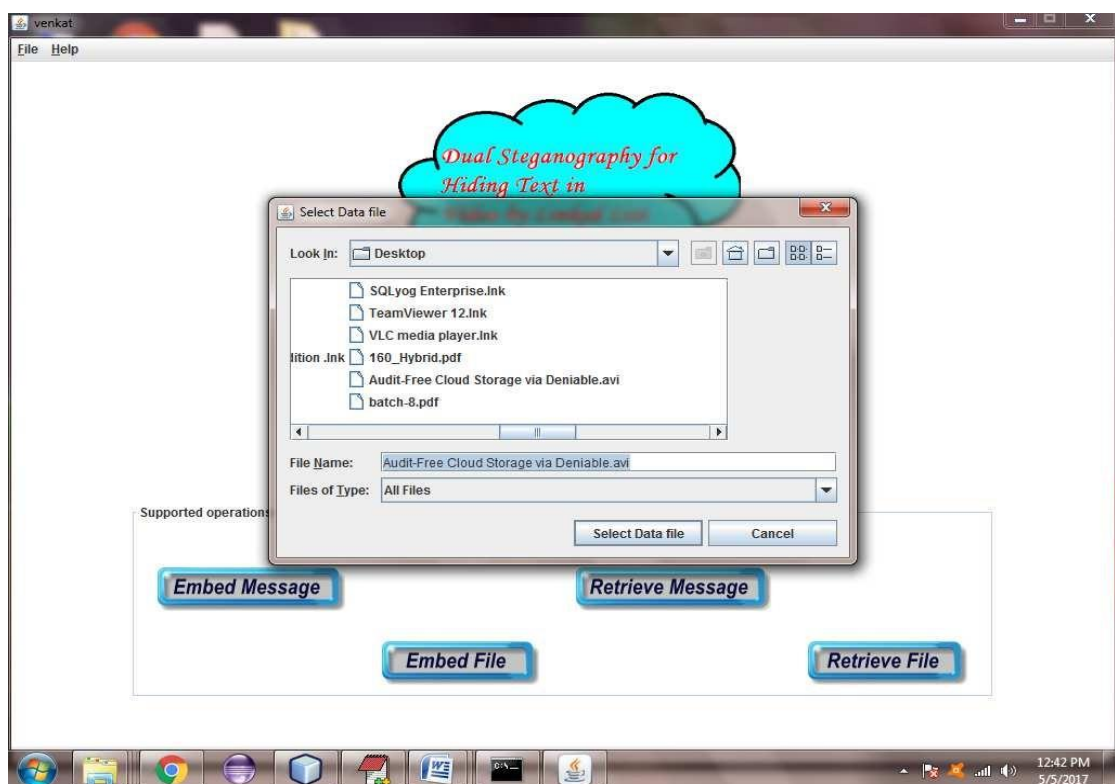
8.1 Home page



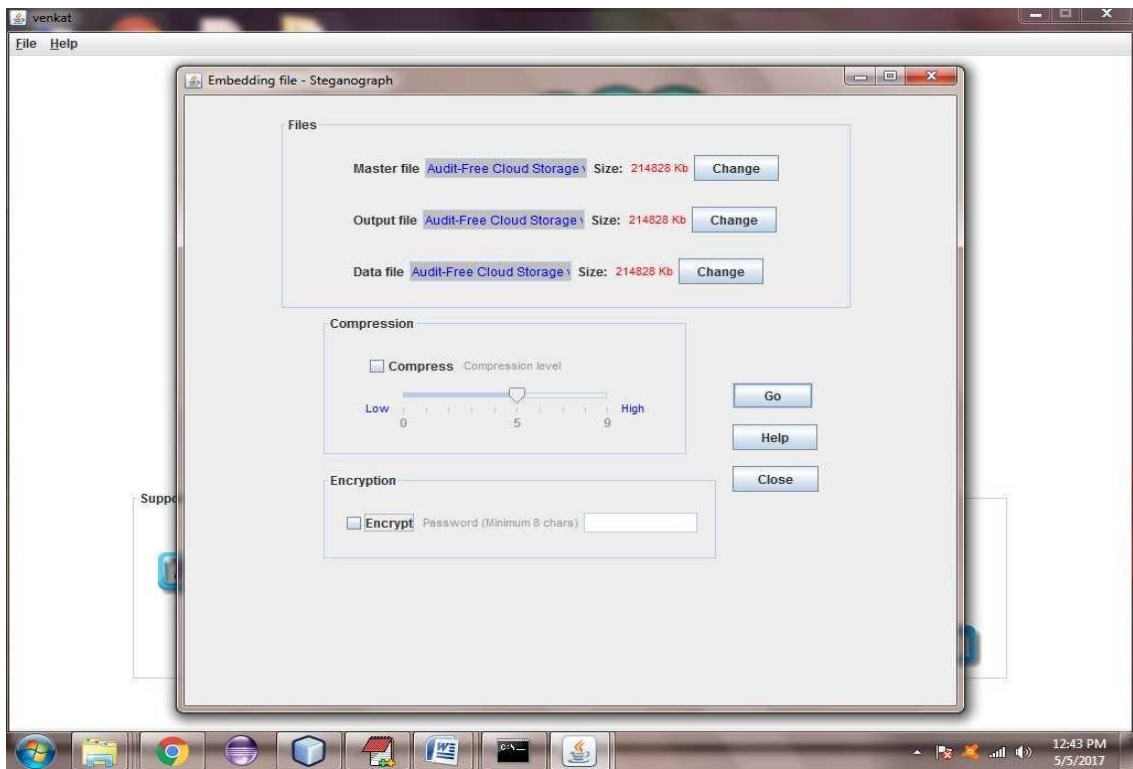
8.2 Embedding File: selecting master video file



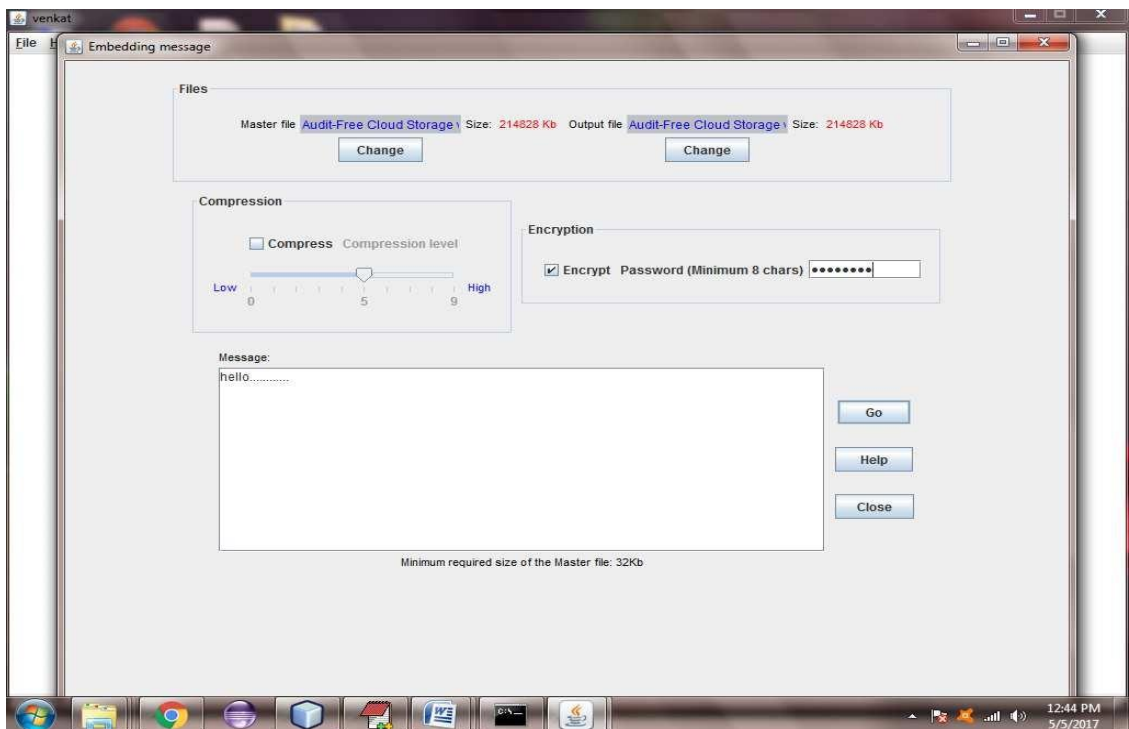
8.3 Embedding file: Selecting Output video file



8.4 Selecting Data file



8.5 Embedding File



8.6 Embedding message by giving password

9. CONCLUSION

The Linked List framework and Feistel Network has been presented for covering Information inside Video. The two rule estimations utilized for information encryption and information embeddings are Feistel Network and Linked List framework autonomously. The work starts with expelling plots from cover video. By then the encryption of information happens utilizing Feistel Network. After encryption of information, the blended information is implanted inside every video follows utilizing Linked List system and Stego graphs are passed on. Along these lines, the Stego Frames are joined to get a Stego Video. This procedure gives an odd state security to the data and the possibility of stego video will be relative to the cover video. Since Feistel Network is utilized for scrambling information, it will be troublesome for the gatecrashers to translate the data.

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Great Teachers are worth more than thousand books, i have them in Our Department

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